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Oefeningenreeks 3C nr. 9

Gegeven: $f'(x) = \gamma_0 + \gamma_1 x + \gamma_2 x^2 + \ ... \ + \gamma_n x^n$

$f(x) = \gamma_0 x + \frac{\gamma_1}{2}x^2 + \frac{\gamma_2}{3}x^3 + \ ... \ + \frac{\gamma_n}{n+1}x^{n+1}$

$f(0) = \gamma_0(0) + \frac{\gamma_1}{2}(0)^2 + \frac{\gamma_2}{3}(0)^3 + \ ... \ + \frac{\gamma_n}{n+1}(0)^{n+1}$

$= 0$

$f(1) = \gamma_0(1) + \frac{\gamma_1}{2}(1)^2 + \frac{\gamma_2}{3}(1)^3 + \ ... \ + \frac{\gamma_n}{n+1}(1)^{n+1}$

$= \gamma_0 + \frac{\gamma_1}{2} + \frac{\gamma_2}{3} + \ ... \ + \frac{\gamma_n}{n+1}$

$= \sum_{i=0}^{n} \frac{\gamma_i}{i+1} \rightarrow$ gegeven $f(1) = 0$ dus $f(0) = f(1) = 0$

$\Rightarrow$ Stelling van Rolle: $f'(c) = 0$ dit geldt.

dus: $\gamma_0 + \gamma_1 c + \gamma_2 c^2 + \ ... \ + \gamma_n c^n = 0$

■

References